MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 5837

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, separate cavity, pulse and CW operation

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ik	Ehk	Pi	tk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	Vdc	W	sec	ft
Maximum:	6.9	+20	350	350	-700	29	+45	12	---	10,000
Minimum:	5.7	-150	---	---	(Note 1)	---	-45	---	120	---

PHYSICAL CHARACTERISTICS:

See figure 1
Cathode: Coated unipotential

Base: See figure 1
Cap: C1-3

TEST CONDITIONS:

Parameter:	Ef	Ec1	Ec2/Ec3	Er
Unit:	V	Vdc	Vdc	Vdc
Test condition 1:	6.3	10	325	Adj for max Po
Test condition 2:	6.3	10	300	Adj for max Po

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4318	Jitter (1)	4, 8	1	egy = 18 to 22 v; Ec1 = -10 Vdc; prr = 1,000; tp = 1.0 µs	t	---	0.6	µs
4318	Jitter (2)	4, 11	2	egy = 18 to 22 v; Ec1 = -10 Vdc; prr = 1,000; tp = 1.0 µs	t	---	0.6	µs
4318	Jitter (3)	4, 7	2	egy = 18 to 22 v; Ec1 = -10 Vdc; prr = 1,000; tp = 1.0 µs	t	---	0.6	µs
	Quality conformance inspection, part 1	12	-					
4229	Total reflector current (1)	6	-	Ec1 = 0; Ec2 = Ec3 = 300 Vdc; Er = -90 Vdc	Ir	---	2.0	µAdc
4289	Heater current	-	-		If	600	750	mA
4213	Reflector voltage (1)	8	1		Er	-30	-70	Vdc
4213	Reflector voltage (2)	9	1		Er	-450	-550	Vdc
4213	Reflector voltage (3)	11	2		Er	-20	-60	Vdc
4213	Reflector voltage (4)	7	2		Er	-285	-365	Vdc
4250	Power output (1)	8	1		Po	50	---	mW
4250	Power output (2)	9	1		Po	50	---	mW
4250	Power output (3)	11	2		Po	10	---	mW
4250	Power output (4)	7	2		Po	20	---	mW
1266	Grid current (1)	8	1		Ic1	0	7	mAdc
1266	Grid current (2)	9	1		Ic1	0	7	mAdc
1266	Grid current (3)	11	2		Ic1	0	7	mAdc
1266	Grid current (4)	7	2		Ic1	0	7	mAdc
1256	Cathode current	9	1		Ik	13	29	mAdc
4214	Cathode emission	9, 10	1		$\frac{\Delta Ik}{Ik}$	---	15	%
1266	Grid leakage	-	-	Ec1 = -90 Vdc; Ef = 6.3 V; no other voltages applied	Ic1-k	---	60	µAdc
4275	Cutoff current	-	-	Ec2 = Ec3 = 325 Vdc; Er = 0; Ec1 = -10 Vdc	Ihk	---	100	µAdc
	Quality conformance inspection, part 2							
1101	Secureness of base, cap, or insert	5	-		---	---	---	---
1031	Low-frequency vibration	-	-	No voltages applied	---	---	---	---
4229	Total reflector current (2)	8	1		Ir	---	50	µAdc

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 - Continued							
4229	Total reflector current (3)	11	2		Ir	---	50	µAdc
4270	Heater - cathode leakage	-	-	Ehk = ± 45 Vdc	Ihk	---	75	µAdc
4280	Electronic tuning range	3, 11	2		ΔF	8	---	MHz
4318	Jitter (4)	4	1	egy = 18 to 22 v; Ecl = -10 Vdc; prr = 1,000; tp = 1.0 µs	t	---	0.6	µs
	Quality conformance inspection, part 3							
---	Life test	2	1	Group C	t	500	---	hrs
	Life-test end points:							
4250	Power output (3)	11	2		Po	8	---	mW
4250	Power output (4)	7	2		Po	16	---	mW

NOTES:

1. Reflector voltage shall never be positive.
2. Tube shall be tested with 1,000 MHz test cavity as specified on Drawing 166-JAN. Frequency approximately 1,000 MHz. Tube operated in N = 1 mode (1-3/4 cycles for complete transit). Adjust ER for maximum power output. Adjust output coupling loop for maximum power output.
3. Vary Er for half-power point each side of maximum. Read frequency at the half-power points.
4. The pulse input shall be between 18 and 22 volts and shall be square. Jitter is the time interval of the output pulse during which the tube exhibits erratic starting.
5. Water-immersion test omitted. Torque shall be applied between upper disk (g3) and molded portion of base and shall be 20 inch-pounds.
6. Static test; tube not oscillating.
7. Tube shall be tested with 2,750 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 2,569.2 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
8. Tube shall be tested with 800 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 800.5 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
9. Tube shall be tested with 2,200 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 2,113.8 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
10. Reduce Ef to 5.7 V while tube is oscillating. The percent dropoff in power output shall be less than the limit shown.
11. Tube shall be tested with 1,150 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 1,130.6 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
12. The AQL for the combined defectives in quality conformance inspection, part 1, shall be 1.0, inspection level II.
13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Air Force - 11, 80
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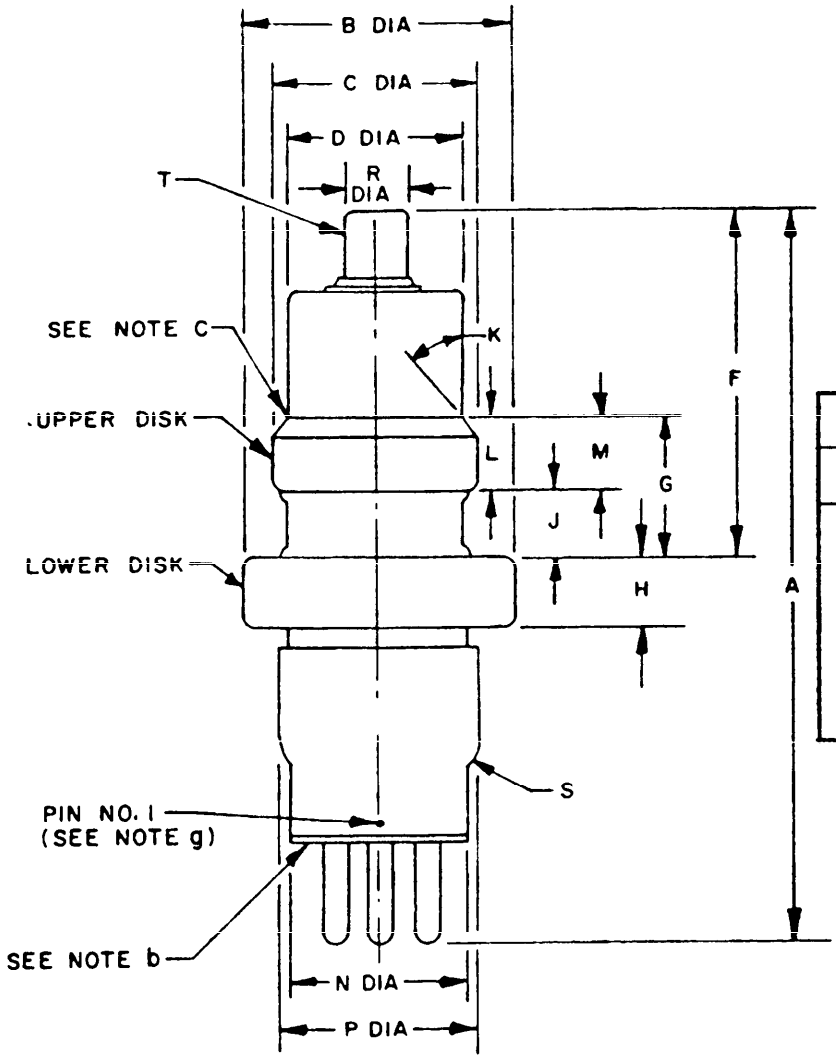
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(Project 5960-2875)

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Pin connections	
Pin No.	Element
1	g^1
2	h
3	k
4	h
Lower disk	g^2
Upper disk	g^3
Cap	r

FIGURE 1. Outline drawing of electron tube type 5837.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
	Qualification	
D	---	.750 (19.05)
K	25°	35°
N	.635 (16.13)	.660 (16.76)
P	---	.750 (19.05)
R	.245 (6.22)	.255 (6.48)
S	Max radius .015 (.38)	
T	Cap: C1-3	
Quality conformance inspection, part 1		
A	2.969 (75.41)	3.219 (81.76)
B	.990 (25.15)	1.000 (25.40)
C	.880 (22.35)	.890 (22.61)
F	1.375 (34.93)	1.500 (38.10)
M	.303 (7.70)	.323 (8.20)
Quality conformance inspection, part 2		
G	.560 (14.22)	.610 (15.49)
H	.250 (6.35)	.312 (7.92)
J	.257 (6.53)	.287 (7.29)
Nominal dimensions		
L	.250 (6.35)	

NOTES:

a. Pin alinement shall meet requirements of JEDEC gage GA4-5.
b. Base shell material to be .010 (.25 mm) nickel, or approved equivalent.
c. Space between glass and upper disk edge or backing ring shall be .030 (.76 mm) maximum. Quality conformance inspection, part 2, shall apply.
d. Diameters B and C shall be concentric within .025 (.64 mm) TIR and shall form circular cylinder within a range of .006 (.15 mm). Allowable radius at ends of cylinder shall be .030 (.76 mm) maximum.
e. Tube shall be tested in concentricity gage specified on Drawing 199-JAN. Gage readings on diameters shall be as follows:

Diameter	Concentricity
D	.050 (1.27 mm) max
N	.060 (1.52 mm) max
P	.060 (1.52 mm) max
R	.040 (1.02 mm) max

f. All concentricity measurements shall be quality conformance inspection, part 2.
g. Pin No. 1 shall be identified by an index mark (colored dot or other).

FIGURE 1. Outline drawing of electron tube type 5837 - Continued.

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